

EE016-3-3-PP1 Project Phase 1 – Investigation in Engineering Project

Chapter 1 : Introduction to the Study

Topic Learning Outcomes

At the end of this topic, you should be able to:

1. Interpret the research problems and the research objectives with justification for the research fostering social awareness and an inclusion of at least Eight United Nations Sustainable Development Goals (UN-SDGs). (C3, PLO2)

Contents & Structure

- Introduction
- Research Problem
- Aim Objectives
- Justification for this Research
- Organisation of the rest of the chapters
- Summary

Recap From Last Lesson

- Nil

Key Terms You Must Be Able To Use

If you have mastered this topic, you should be able to use the following terms correctly in your assessments:

- Research Problem
- Aim
- Objectives
- Justification for this Research

CHAPTER 1 – INTRODUCTION OF THE STUDY

Outline

- i. Introduction
- ii. Research Problem
- iii. Aim and Objectives
- iv. Justification for this Research
- v. Organization of the rest of the chapters
- vi. Summary

CHAPTER 1 – INTRODUCTION OF THE STUDY

1.1 Introduction

- Give a brief background about the project with proper citation.

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1.2 Research Problem

Research Problem

- Identify the current engineering problem with proper citation and / or the gap left over to be solved in the existing system.
- The solution to this problem must be solved in the proposed project by fostering social awareness and integrating at least eight UN-SDGs.
- The problems stated must not be general problem with wider scope and hence, it must be narrowed to what the proposed project is intended to solve.

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1.2 Research Problem

Research Problem

- A research problem can be framed around social awareness if:
 - The issue involves human behaviour, community engagement, public understanding, or societal impact.
 - The goal is to improve equity, sustainability, or collective well-being.
 - The research aims to address social challenges with evidence-based insights.

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1.2 Research Problem

Research Problem

Why integrating SDGs?

- You may integrate 8+ SDGs if your research touches multiple interconnected global issues such as:
 - health (SDG 3)
 - education (SDG 4)
 - gender equality (SDG 5)
 - economic development (SDG 8)
 - reducing inequalities (SDG 10)
 - sustainable communities (SDG 11)
 - strong institutions (SDG 16)
 - partnerships (SDG 17)
- Many real-world problems are interdisciplinary, so aligning the problem with multiple SDGs is academically sound and globally relevant.

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1.2 Research Problem

Research Problem

Sample Research Problem (Engineering – Undergraduate Level)

“Despite increasing efforts toward sustainable development, many urban communities still lack accessible, affordable, and user-friendly technological solutions that promote social awareness regarding energy consumption, waste management, and environmental responsibility. Existing systems are often too technical, fragmented, or poorly adopted, leading to low public engagement and limited behavioural change. This creates a gap in developing integrated engineering solutions that simultaneously address key UN-SDGs—including SDG 3 (Good Health), 4 (Quality Education), 5 (Gender Equality), 7 (Affordable and Clean Energy), 10 (Reduced Inequalities), 11 (Sustainable Cities), 12 (Responsible Consumption), and 13 (Climate Action). Therefore, an engineering challenge arises: How can a technology-driven system be designed to effectively foster social awareness while supporting multiple SDGs in a scalable and community-friendly manner?”

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1.3 Aim and Objectives

Aim and objectives must be correctly and clearly aligned with complex engineering problems, fostering social awareness and integrating at least eight UN-SDGs.

Aim:

- Aim describes the purpose and intention for undertaking the particular project proposed.
- Aim is about what you hope to do, your overall intention in the project. It signals what and/or where you aspire to be by the end. It is what you want to know.
- It is the point of doing the research.
- An aim is therefore generally broad. It is ambitious, but not beyond possibility.

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1.3 Aim and Objectives

Objectives:

- Objectives relate to the expected outcomes of the project.
- You should break these down into 'primary objectives' which you guarantee to achieve and 'extensions' which will only be implemented if time allows.
- The primary objectives should be clearly specified, and must be minimum of three objectives with the Blooms Taxonomy of Cognitive level 5/6.
- But the extensions may be vaguer.

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Blooms Taxonomy of Cognitive level 5/6 – Keywords

- Design
- Develop
- Simulate
- Evaluate
- Examine
- Compare
- Contrast
- Justify
- Measure
- Build
- Construct
- Discuss
- Formulate
- Propose
- Modify

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1.3 Aim and Objectives

Aim (Example)

- The aim of the project was to develop an improved patient monitoring system using steganography, fostering social awareness and integrating at least eight United Nations Sustainable Development Goals (UN-SDGs), particularly those related to good health and well-being, innovation, reduced inequalities, sustainable communities, and responsible resource use.

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1.3 Aim and Objectives

Objectives (Examples)

- To develop a sensing and data conditioning system in order to acquire patient's vital body parameters, exemplifying social awareness and aligning with UN-SDGs such as Good Health and Well-Being (SDG 3) and Industry, Innovation, and Infrastructure (SDG 9).
- To develop an algorithm using steganography to transmit these acquired information through a secured channel, thereby fostering inclusivity, privacy, and security in healthcare systems, while supporting Reduced Inequalities (SDG 10) and Peace, Justice, and Strong Institutions (SDG 16).
- To evaluate the performance of the developed system by considering efficiency, accessibility, and sustainability, integrating Responsible Consumption and Production (SDG 12), Sustainable Cities and Communities (SDG 11), Climate Action (SDG 13), and Partnerships for the Goals (SDG 17).

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1.4 Justification for this Research

- You are expected to provide an explanation about why the research is necessary, fostering social awareness and integrating at least eight UN-SDGs.
- A rationale is to be constructed and may illustrate how the research endeavour addresses gaps in the existing knowledge base, contributes a new dimension or perspective, or generates theory about a phenomenon that has not been explored previously.

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1.5 Organization of the rest of the chapters

- Brief the readers on what is in the next two chapters.

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Examples

- Chapter 2 reviews the work done by other researches in the data conditioning system developed and the algorithms using steganography, problem statement overcome by them along with the methodology and outcomes achieved by them.
- Chapter 3 proposes the methodology to be implemented in achieving the aim and objective of this project. It also deals with the investigation of the components and tools used along with a proper justification for the same. The design specifications are derived from the literature review and the design is obtained by deriving from the fundamental principles. Also, Professional engineering practices and project management of this project are explained.

CHAPTER 1 – INTRODUCTION OF THE STUDY

1.6 Summary

- Summarize the entire chapter.

QUICK REVIEW QUESTIONS

- What is a research problem?
- How many aim must a project have?
- How many objectives must a project have?

SUMMARY OF MAIN TEACHING POINTS

- Introduction to the study
- Research Problem
- Aim
- Objectives
- Justification for this Research

QUESTION & ANSWER SESSION

Q & A

WHAT WE WILL COVER NEXT

- Literature Review